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Remaining Useful Life Prognosis with Applications

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Abstract

Remaining Useful Life (RUL) prediction is central to maintenance planning in the manufacturing sector. It refers to the study of when some specified equipment or system is likely to fail, based on its current operating characteristics and therefore allows engineers to schedule maintenance, avoid unplanned disruptions and streamline operations, thereby reducing costs and enhancing efficiency. In this paper, we present several data-driven applications for system health monitoring with a focus on oilfield applications, such as artificial lift systems.

A holistic and reliably predictive view of RUL requires addressing the effect of various factors on system degradation, such as environment variables and the prior history of operation, both of which can run into thousands of variables with often confounding effects. We describe one such approach for RUL prediction that can potentially highlight causative factors for equipment failure and degradation, that is known as survival analyses. These approaches are paired with workflows that assess the effect of uncertainty on the predictions of equipment health and RUL.

We demonstrate the application of survival analyses with a presentation of the prognostic or causative factors driving system degradation. Our applications specifically include field-wide RUL analyses for artificial lift systems, including ESPs. For problems such as these, model-based RUL prediction becomes challenging simply because the formulation of a system model with the complexities in the underlying physics becomes intractable. This is where appropriate feature engineering is critical.

Our work presents an argument for increased adoption of RUL prediction methods in the petroleum industry. Failures and unwarranted downtime are commonplace, and the methods presented here have the advantage of being employed by even smaller, regional operators and service companies. To our knowledge, this is the first discussion of Remaining Useful Life prediction methods combined with feature engineering.

Introduction

Survival analysis, also known as time-to-event analysis, measures the time until an event takes place (Hougaard, 2000). In the context of ESPs, survival time analysis facilitates various operational tasks, such as comparison of pumps from various manufacturers or comparison of median survival time for pumps deployed in different fields with different fluid types, and causal inference for failures. These failures are

commonly of three types: electrical, mechanical, and operational (Fakher et al. 2021). There have been a few prior field case studies that showcase the utility of survival analyses. An early study by Divine et al. (1993) described failure modes, pump mortality concept, and pump reliability of 135 stage pumps produced in West Texas. A later work by Bailey et al. (2006) included 177 ESP systems with 64 associated variables. They used Kaplan-Meier and log-rank tests to identify the most significant factors governing pump reliability. Camilleri et al. (2012) determined the run life of 22 ESP installations on 12 producing wells in offshore Gabon. Mean time between failure (MTBF) and Survivability fraction, an exponential model based on MTBF, were calculated to analyze the ESP performance. An MTBF of 2.44 years was obtained. A quantitative analysis of 10 failures indicated that the cable issues and corrosion were the primary causes of failure.

Camacho et al. (2016) performed a survival analysis using data from 10 years of ESPs installed in a North Sea field. The metrics chosen were the mean time to failure (MTTF) and the creation of a parametric model based on Weibull distribution. According to the survivability plots, 88% of the installed ESPs survived more than 90 days, and less than 35% of them survived more than three years. The failures detected were related to equipment defects, incorrect pump installation, and pump operation problems, including a lack of continuous monitoring. A new high-reliability ESP system was created based on improvements in ESP transportation, zero-defect manufacturing, and 24x7 Surveillance. Five new ESPs were installed, all surviving over 90 days, eliminating premature failure. By the time the study was published, the five new ESPs were operating without failure with a cumulative run-life of more than 2,170 days.

Pastre & Fastovets (2017) applied survival analysis to measure ESPs failure rates in early and long-life periods. The ESPs population consisted of 219 ESP installations and 162 failures at the central North Sea. The ESP components were classified into two subsystems: mechanical and electrical. Based on the most typical failures, an improvement of the components in each category was performed. The metrics used to evaluate the short- and long-term enhancement were the mean time to failure (MTTF), Kaplan-Meier estimator, and a parametric model based on Weibull distribution. Three generations of the mechanical and electrical subsystems were developed and monitored for seven years. Kaplan-Meier and Weibull curves were used to compare the performance of each generation to quantify the increase in reliability of each subsystem after each improvement.

Villalobos Leon et al. (2018) conducted an analysis of real-time field data to determine the optimal configurations for ESPs design in order to enhance their performance. The survival parametric models used were the mean time between failures (MTBF) and the probability of failure. After implementing the improvement strategies, the survival probability increased by 66% from year 1 to year 3, and the rate of failures occurring in less than 90 days decreased by 86% from the first to the third year.

Mogollon et al. (2018) aimed to increase the mean time between failure (MTBF) of the ESPs in the Rubiales field in Colombia. During five years (from 2008 to 2012), data was collected on the performance of the installed ESPs, repetitive causes of failures, and examples of how they were solved. Based on the lessons learned, actions were taken to improve the most frequently failing ESP components and standardized pump selection parameters for the five different types of wells present in the field. After the improvements, the MTBF of the pumps installed in the field increased from 1,581 days to 3,152 days, demonstrating that it is possible to increase the lifetime of the pumps by taking appropriate actions.

Al-Jazzaf et al. (2019) developed a methodology that combines Mean Time to Failure (MTTF), Dynamic Average Equipment Run Life and Weibull method prediction distribution of failures over time to construct ESP survival curves based on operating performance, reservoir data, and design and equipment information. The dataset comprises historical data from over eight years of 5,000 ESP systems installed in Kuwait. Clusters of wells were identified and compared with each other. This information allowed to identify the wells with poor pump performance and implement strategies to significantly improve the lifetime of the ESPs located in those wells.

[Khabibullin et al. \(2020\)](#) evaluated the failure time of ESPs using high-frequency data from 5 - 7 years in a field in Western Siberia. The parameters used were Mean time between failures (MTBF) and survival curves based on a parametric distribution function. 50% of the pumps had a time of 400 days before failure. The ESPs were divided into three categories according to their depth: pumps with a depth less than 1000 m, a depth greater than 1000 m, and a depth greater than 2000 m. The survival curves for each of the three categories were compared. The wells with ESPs at depths greater than 2000 m had the shortest life span, and those with ESPs installed at depths less than 1000 m survived the longest.

[Sutyryna et al. \(2022\)](#) counted on a dataset of approximately 23,000 rows (ESPs) and 130 variables, including pump status (failure or non-failure) and lifetime. First, the Mean time between failure and average run life (ARL) was calculated. Then, the significance of each variable was determined by calculating the p-value. Kaplan-Meier curves of the three motor types were performed. After one year, the type 2 motor has a survival probability of 81% compared to 69% and 60% for pumps with type 1 and type 3 motors, respectively. Similarly, Kaplan-Meier curves of different pump configurations under the same operating conditions were compared to determine the parameters (equipment configuration data and downhole conditions) with the most significant influence on ESP survival. These parameters are pump stage thrust type, cable temperature rating, and cable insulation type.

In this study, we first describe RUL (remaining-useful-life) predictions using survival analyses. We then describe its application to a field case study of ESP pumps operating in offshore Ecuador. In addition to the RUL analyses, we demonstrate leveraging feature engineering to gain additional insights into pump failure and identify key features critical for pump health.

Survival Analyses: Description and Mathematical Formulation

Censoring

It is important to point out that in the majority of survival studies, not all individuals start participating simultaneously ([Cantor, 2003](#)). In order to correctly collect a time to event, it is necessary to establish an unambiguous time origin, define the event of interest (when the "clock stops"), and set an appropriate time scale ([Hosmer et al., 2008](#)).

Survival data are characterized by a mixture of complete and incomplete observations ([Aalen et al., 2008](#)). Censoring is the term used to indicate when the exact time to event for an included observation is unknown, and the observation is classified as being censored ([Hosmer et al., 2008](#)). Censoring can happen due to different factors, including the end of the study prior to the occurrence of the event, the study is out of time or resources to be completed, or loss of follow-up during the study period, or an individual drops out from the study because of illness or death, and failure of equipment before the next control date ([Kleinbaum & Klein, 2012](#)). Survival censoring data can be classified into three types: right censored, left censored, and interval censored.

Definitions for Survival Analyses

Let T be the random variable that represents the survival time. The probability that survival time will be equal to or less than time, t is $F(t)$. The survival function $S(t)$, also known as the reliability function in the context of equipment failures, defines the proportion/probability of individuals that have not failed till time, t as shown in Equation 1 ([Cantor, 2003](#); [Hosmer et al., 2008](#)):

$$S(t) = Pr(T > t) \quad (1)$$

Note that as time, t increases, the survival function tends toward zero because, over time, an increasing number of individuals may experience the event of interest (i.e. not survive or fail), which leads to a reduction of the probability that it has not occurred by that time ([Aalen et al., 2008](#)). The hazard function h

(t) is the frequency of events (or failures) per unit time at time, t and is denoted as Equation 2 (Kartsonaki, 2016).

$$h(t) = \lim_{\delta t \rightarrow 0} \frac{S(t) - S(t + \delta t)}{\delta t} = -\frac{dS(t)}{dt} \quad (2)$$

Different survival curve types can be found, but all of them are nonincreasing (monotonically decreases), starting at one when time is zero and are equal to zero as time approaches infinity (Fig 1.).

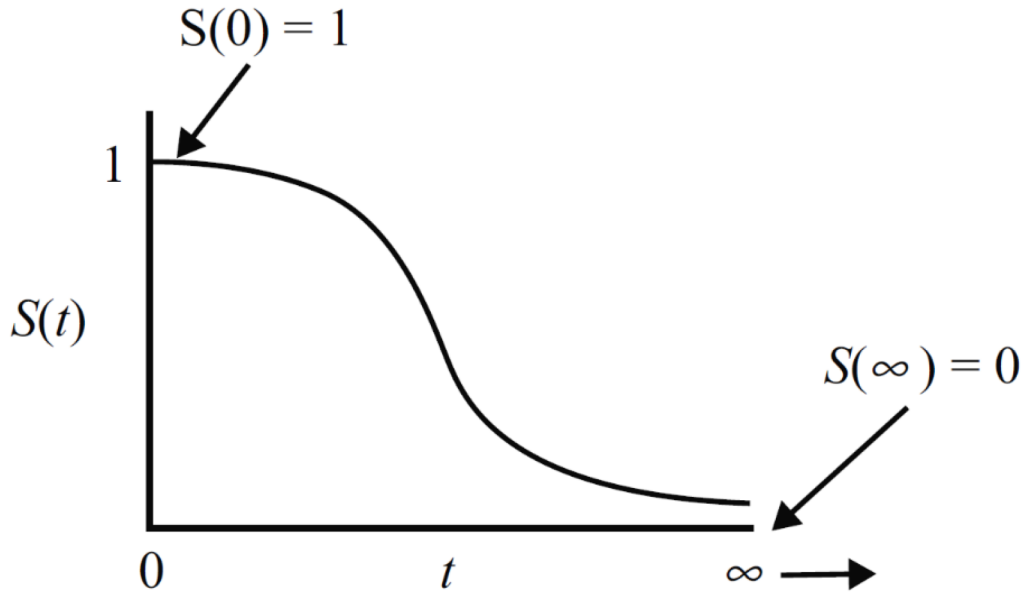


Figure 1—Theoretical survival curve (Kleinbaum & Klein, 2012)

Kaplan-Meier method

This method was proposed by Kaplan & Meier (1958) and is a non-parametric method to estimate the survival curves. This technique integrates data from all available observations, including uncensored and censored survival times, by considering survival at any given time as a sequence of discrete steps determined by observed survival and censored times (Hosmer et al., 2008).

Let the number of individuals who have an event at time $t_{(i)}$ be denoted d_i and the number of observations at risk at $t_{(i)}$ be n_i . The Kaplan-Meier estimator of the survival function at time t is as Equation 3 (Klein & Moeschberger, 2003):

$$\hat{S}(t) = \prod_{t_{(i)} \leq t} \left(\frac{n_i - d_i}{n_i} \right), \quad \begin{matrix} \text{if } t < t_{(i)} \\ \text{if } t_{(i)} \leq t \end{matrix} \quad (3)$$

Non-parametric Analysis of Survival Data

Next, we discuss two common non-parametric methods to summarize survival data and facilitate comparisons.

Log-rank test

The log-rank test is used to compare the survival distribution between groups defined by categorical covariates. Based on the assumption that there is no difference between the groups, it compares the observed number of events at each time point with the expected number of events. A chi-square test is calculated based on the difference between the observed and expected number of events. For the log-rank test, censoring is non-informative, which means that the probability of being censored is unrelated to survival times.

Additionally, it assumes that hazard rates for the groups being compared remain proportional over time, which means that hazard functions remain constant (Cantor, 2003; Kleinbaum & Klein, 2012; Marques de Sa, 2007).

The null hypothesis (H_0) of the test is that the survival curves of the compared groups are the same. The alternative hypothesis (H_1), also known as the rejection of the null hypothesis, suggests a significant difference in the survival distributions between the groups. P-value for the log-rank test is determined from tables of the chi-square distribution with one degree of freedom. If the p-value is greater than α significance level, the null hypothesis is accepted (Cantor, 2003; Kleinbaum & Klein, 2012; Sestelo, 2017).

Survival distributions are calculated using Equation 4. For example, we may want to compute the differences in survival time for two different values of a specific variable. More specifically, we may want to see if survivability of a pump is different for high and low values of fluid flow rates. We can construct a log-rank test as shown in Table 1.

Table 1—An example of a hypothesis test constructed to test if survivability of a pump is different for different values of flow rates.

Null hypothesis, H_0	The survival curves for the high and low values of the fluid flow rates are equal.
Alternative hypothesis, H_1	The survival curves associated with pumps operating at high and low flow rates are significantly different.

Semi-parametric Cox Proportional Hazards Model

The major contribution of the paper by Cox (1972) was to introduce a method for estimating the hazard (h) at time t as the product of two quantities, the baseline hazard function and the explanatory variables (Hougaard, 2000; Kleinbaum & Klein, 2012). The Cox proportional hazards model has the form of Equation 4 and is considered to be fairly robust irrespective of the actual parametric form:

$$h(t, X) = h_0(t) \cdot e^{\sum_{i=1}^p \beta_i X_i} \quad (4)$$

Where $X=(X_1, X_2, \dots, X_p)$ and represents the explanatory or predictor variables, $h_0(t)$ is the baseline hazard and it is a function of time, $e^{\sum_{i=1}^p \beta_i X_i}$ is the linear sum of $\beta_i X_i$ over the p explanatory time independent variables, and β_i is a coefficient that describes the effect of the predictor variables (Aalen et al., 2008; Kleinbaum & Klein, 2012).

Note that if all the explanatory variables are equal to zero, the Cox expression is simplified to the baseline hazard function since the exponential term of the equation reduce to a value of one. The Cox model is considered semiparametric because the baseline hazard is an unspecified function treated nonparametrically, and a parametric form is only assumed for the effect of the predictor variables, also known as covariates (Klein & Moeschberger, 2003; Kleinbaum & Klein, 2012). In this work, however we are not considering the CPH model for analysis. Table 2 provides an overview of prior work done with survival analyses over the past few years.

Table 2—Previous studies on the application of survival analysis to ESP failure prediction

Authors	Dataset Size	Survival Model Applied	Study Objective
Divine et al. (1993)	135 stage pumps producing from West Texas.	Parametric model: Mean time between failure (MTBF).	Determine pumps condition and expected life.
Bailey et al. (2006)	177 ESP systems with 64 variables.	Non-parametric: Kaplan-Meier and Log-rank test. Semi-parametric: Cox proportional hazards model.	Developed a methodology for understanding and predicting the behavior of ESPs by considering variables that may impact their performance.
Camilleri et al. (2012)	12 producing wells real-time ESP data with 22 ESP installations in Gabon. High-frequency data	Parametric model: Mean time between failure (MTBF), and Survivability fraction.	Determine ESPs run life.

Authors	Dataset Size	Survival Model Applied	Study Objective
Camacho et al. (2016)	ESP performance data for 10 years of operations in the North Sea. 5 high-reliability ESP systems data for 20 months.	Parametric model: Mean time between failure (MTBF), and Weibull cumulative distribution function.	Estimate the probability of an ESP system surviving a certain amount of time. Create a reliability model for the new high-reliability ESPs.
Pastre & Fastovets (2017)	219 ESP installations and 162 failures at the central North Sea.	Parametric: Mean time between failure (MTBF), and Weibull cumulative distribution function. Non-parametric: Kaplan-Meier.	Measure ESPs failure rates in early and long life periods.
Villalobos Leon et al. (2018)	288 ESP operations: 32 new installations, 128 installation workover and 128 pulls.	Parametric model: Mean time between failure (MTBF), and Survivability fraction.	Achieve the expected run life and continuously improve the MTBF. Minimize the failure probability.
Mogollon et al. (2018)	550 ESP performance data from over 10 years in Rubiales Field in Colombia.	Parametric model: Mean time between failure (MTBF), and Survivability fraction.	Determine ESPs run life. Increase the mean time between failure (MTBF)
Al-Jazzaf et al. (2019)	5,000 ESP performance data from over 8 years in Kuwait.	Parametric model: Mean Time to Failure (MTTF), Dynamic Average Equipment Run Life, and Weibull distribution function of failures over time	Interpretation of ESP Reliability using Weibull Analysis and Dynamic Run Life Analysis.
Khabibullin et al. (2020)	ESP high-frequency data from 5 - 7 years in a field in Western Siberia.	Parametric model: Mean Time to Failure (MTTF) and survival curves based on parametric distribution function.	Construction of reliability curves of ESPs and determine their confidence intervals.
Sutyryna et al. (2022)	23,000 rows (ESPs) and 130 input parameters.	Parametric: Average run life (ARL), and Mean time between failure (MTBF). Non-parametric: Kaplan-Meier. Semi-parametric: Cox proportional hazards model.	Determine the number of failures and the run life of ESPs.

Description of the Data Used in this Study

The data are publicly available and published by SPE Ecuador section in 2021 ([SPE Ecuador, 2021](#)). The dataset is spread over several files and contains high frequency data and production data for 99 unique deepwater wells that have experienced pump shutdowns. These shutdowns are because of either a failure in the pump or a manual shutdown (not a failure). The variables that describe the data include:

High Frequency Data

- CURRENT (Amperes)
- PRESS_DESC: Discharge Pressure (psi)
- FREQUENCY (Hz)
- PRES_INT: Intake Pressure (psi)
- TEMP_INT: Intake Temperature (°F)
- TEMP_MOT: Engine temperature (°F)
- OUT_VOLT: Output Voltage (V)
- PRES_INTK: Intake Pressure (psi)
- TEMP_INTK: Intake Temperature (°F)
- Vibration: Normal Range Values lie between 0 - 0.8

Production Data

- BFPD: Barrels of Fluid per Day
- BOPD: Barrels of Oil per Day
- BWPD: Barrels of Water per Day
- MSCF: Million of Standard Cubic Feet
- BSW: Basic sediment and water (%)
- GOR (MSFC/BPPD): Gas-oil Relation
- GLR (SCF/BFPD): Gas-liquid Relation
- API: Oil API Gravity
- FREC (Hz): Frequency
- PIP (psi): Pump intake pressure
- PROF INTAKE (ft): Intake Depth
- AMPERAGE: Current
- PUMP: Pump name
- PSI_CAB: Wellhead Pressure

Feature Engineering

This section presents the feature engineering methodology developed to address data limitations and extract meaningful indicators of ESP degradation. The approach focuses on creating two primary engineered features: cumulative sediment volume and frequency-normalized flow rate, both designed to capture different aspects of pump degradation over time.

Data Characteristics and Challenges

The ESP dataset presents several characteristics that necessitate specialized feature engineering approaches:

Temporal Resolution Disparities. The dataset comprises two distinct temporal resolutions: high-frequency operational data (averaging 242 samples per day) covering approximately 17 months, and low-frequency production data (averaging 1.6 samples per month) spanning 72 months. This temporal mismatch presents challenges for feature engineering, as the target failure labels are only available in the high-frequency data, while critical production parameters such as flow rate (BFPD) and sediment content (BSW) are contained in the low-frequency dataset.

Limited Failure Event Coverage. A significant limitation for remaining useful life estimation is that only 4 wells out of 99 contain at least two failure events. This constraint is critical because wells with two failures allow the assumption that between failures, the pump begins with maximum health and deteriorates to minimum health at the subsequent failure. The scarcity of complete pump lifecycles makes it difficult to estimate pump lifetime, particularly since pump installation dates are not available in the dataset. Most wells therefore represent partial lifecycle observations, complicating efforts to establish baseline degradation patterns.

Operational Parameter Inconsistencies. Analysis of the production data reveals inconsistencies between theoretically coupled parameters. Pump curve theory predicts that discharge pressure and flow rates are tightly coupled, with measured data constrained to specific speed lines on the pump performance curve.

These curves consist of concentric speed lines (e.g., 40Hz, 45Hz, 50Hz, 55Hz, 60Hz) centered around the origin, where increasing operating frequency typically increases achievable flow rate. However, the observed data shows poor correlation between flow rate (BFPD, a low-frequency measurement) and discharge pressure (a high-frequency measurement), which should be inversely proportional (See Fig. 2). This inconsistency is attributed to the noisy characteristics of low-frequency measurements and different sampling methodologies between sensor types.

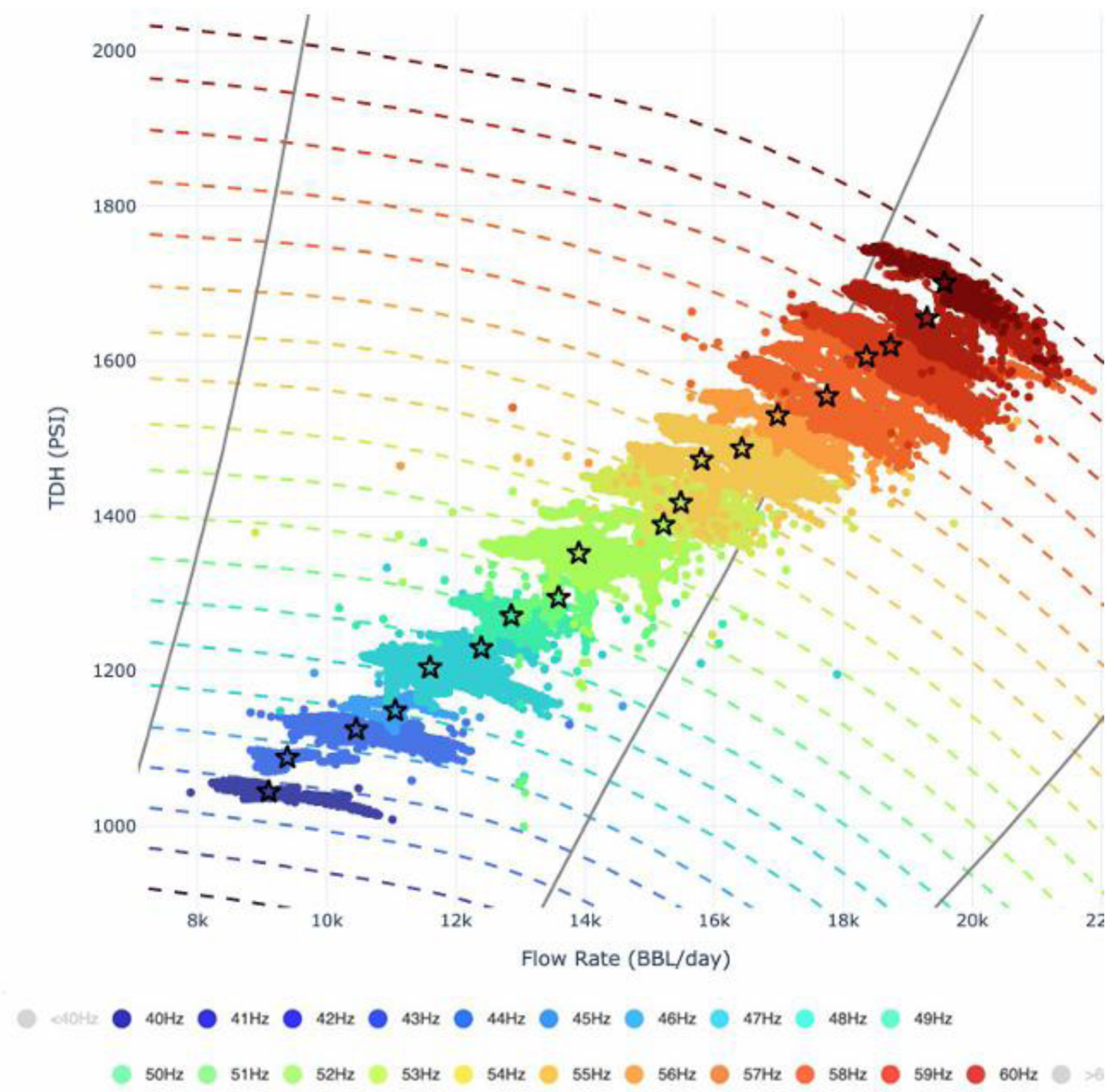


Figure 2—Typical pump curve showing correlations between discharge pressure and flow rate at different operating frequencies. Data points are colored by pump operating frequency. Though, the poor correlation between discharge pressure and flow rate within this paper is an artifact of the dataset studied rather than a generalizable quality of pumps (Adapted from Lewis et al. 2025).

Frequency Synchronization Issues. The operational frequency values present in both high and low-frequency datasets exhibit temporal desynchronization. Low-frequency frequency values appear somewhat random while high-frequency frequency values may gradually increase over time. This inconsistency

complicates efforts to normalize flow rate measurements to account for varying pump speeds, which is essential for isolating degradation-related changes from operational adjustments.

Flow Rate Measurement Considerations. Flow rate measurements are particularly important because they serve dual purposes: they are tightly tied to revenue generation and can function as proxies for both financial and physical system health. Two distinct measurement approaches exist for flow rate quantification:

1. **Aggregate Flow Rate:** The sum of barrels produced within a 24-hour period, representing total daily production volume.
2. **Instantaneous Flow Rate:** A velocity-based measurement representing the flow rate that would be achieved if the pump operated continuously for 24 hours.

These measurements differ significantly when pumps operate intermittently. For example, a pump operating at maximum capacity for 12 hours then shutting off for 12 hours would yield an instantaneous measurement twice the aggregate measurement. For pump health assessment, instantaneous flow rate is more informative as it reflects the pump's actual capacity independent of operational scheduling.

Data Preprocessing and Smoothing

To address the noise characteristics of the low-frequency data, a centered moving average preprocessing step was implemented. A 181-day sliding window was selected to balance noise reduction with degradation signal preservation. This window captures approximately 10 samples on average and provides sufficient temporal context for meaningful trend identification while avoiding excessive smoothing that could mask degradation variations.

The centered approach utilizes 90 days of past and future data to compute the mean, which eliminates lag artifacts inherent in backward-looking moving averages. While the use of future data is generally avoided in predictive modeling contexts, this preprocessing step serves the specific purpose of feature evaluation rather than real-time prediction. The approach simulates a richer dataset with higher sampling rates to assess feature utility. The moving window is truncated when system changes occur, specifically when pump failures are detected within the next 90 days, to prevent contamination across pump replacement events where system characteristics are likely to change substantially.

Engineered Features

Frequency-Normalized Flow Rate Feature. The first engineered feature addresses pump efficiency degradation by normalizing flow rates to a standard operating frequency. This feature is motivated by the observation that as pump systems degrade, operators typically increase pump speeds to maintain target production volumes, masking underlying efficiency losses.

Flow rate degradation can result from multiple mechanisms:

1. **Cavitation:** Operating too far from the best efficiency point on the pump curve causes vacuum bubbles that erode pump impeller surfaces
2. **Scale Buildup:** Mineral deposits in piping reduce effective flow capacity
3. **General Wear:** Mechanical degradation of pump components reduces efficiency

The normalization process addresses the practical challenge that operators cannot easily perform frequency calibration sweeps to establish baseline relationships. Such sweeps would require operating pumps at each frequency (e.g., 40-60Hz) for sufficient time (approximately 2 hours per frequency) to allow system pressure stabilization. However, operational constraints make this impractical:

1. Operators are typically constrained by daily production volume targets
2. Reduced frequency operation would be financially counterproductive

3. Utility providers may impose significant fees based on peak kWh consumption periods
4. Scheduling and coordination overhead creates an operational burden

Instead, the normalization process involves two primary steps:

1. Frequency Offset Learning: Historical frequency transitions are analyzed to determine the relationship between frequency changes and corresponding flow rate adjustments. A simplified approach calculates the mean offset per Hz across all observed frequency transitions:

$$\text{Offset per Hertz} = \text{average} \frac{\delta(\text{Flow rate})}{\delta(\text{Hertz})} \quad (5)$$

2. Flow Rate Normalization: Observed flow rates are adjusted to their equivalent values at a reference frequency (60 Hz):

$$\text{BFPD Normalized to 60Hz} = (\text{BFPD}_{\text{observed}}) + \text{Offset per Hz} * (60 - \text{Frequency observed}) \quad (6)$$

This normalization enables the isolation of well productivity trends from operational pump speed adjustments, theoretically revealing degradation-related flow rate reductions that would otherwise be masked by frequency compensation. The hypothesis is that there exists a common minimum flow rate threshold before system failure, which becomes more apparent when frequency-related variations are removed.

Fig. 3 demonstrates the effectiveness of the flow rate normalization process by comparing the before and after time series. The comparison illustrates how frequency normalization removes operational speed adjustments to reveal underlying degradation trends that would otherwise be obscured by pump speed compensation strategies employed by operators.



Figure 3—Two-panel comparison showing (top) original BFPD flow rate time series with frequency variations and (bottom) frequency-normalized flow rate adjusted to 60Hz equivalent, revealing underlying degradation trends. For instance, the location C in the bottom picture can now be associated with a pump struggling to supply fluid even while operating at moderate flow rates (Adapted from Lewis et al. 2025).

Cumulative Sediment Volume Feature. The second engineered feature addresses the hypothesis that sediment passage through ESP systems contributes to cumulative wear and degradation. This feature targets

gradual abrasive wear that occurs when sediment materials pass through pump components over time, representing one of two primary degradation mechanisms (the other being large impulse forces such as seismic events).

The feature is calculated as:

$$\text{Cumulative Sediment} = \sum_{t=0}^{t=t} \text{BFPD}_{\text{Observed}} * \text{BSW} \quad (7)$$

where BSW represents the Basic Sediment and Water percentage, and BFPD denotes barrels of fluid per day.

The underlying assumption is that each unit volume of sediment contributes approximately equivalent degradation to pump components. If this hypothesis is valid, pumps should exhibit characteristic maximum sediment volumes before failure, creating identifiable thresholds for predictive modeling. While not all pumps will degrade primarily due to high sediment volumes, those that do should show a subset of failures occurring at relatively consistent cumulative sediment levels.

A critical enhancement to this feature involves resetting the cumulative sum following detected pump failures. When a failure event is identified, the cumulative sediment counter resets to zero, reflecting the assumption that pump replacement restores the system to an undegraded state. This reset mechanism prevents the artificial accumulation of sediment values across different pump installations within the same well.

While using target values to derive features might appear to constitute data leakage, this concern applies only to artificial train-test separations rather than real-world applications. In operational streaming scenarios, failure knowledge would be used to reset cumulative features only after failure occurrence, not before. The approach would be appropriate for forecasting problems where test data continuously transitions into training data.

Expected Outcomes and Hypotheses

The engineered features are designed to test specific hypotheses about ESP degradation:

1. **Sediment Degradation Hypothesis:** If cumulative sediment volume is a significant degradation factor, wells should exhibit characteristic maximum sediment thresholds before failure, with reset cumulative values providing superior predictive performance compared to non-reset versions.
2. **Efficiency Degradation Hypothesis:** If pump efficiency degradation is predictive of failure, frequency-normalized flow rates should demonstrate clearer degradation trends compared to raw flow rate measurements, enabling earlier failure prediction. The minimum flow rate threshold concept should be supported by data analysis.
3. **Feature Validity Hypothesis:** If the engineered features capture genuine physical relationships rather than spurious correlations, models trained on real data should significantly outperform those trained on shuffled control datasets. While we do not show those results here in the interests of brevity, in our study, the engineered features showed excellent correspondence with pump health in comparison to a separate scenario with randomly shuffled data.

The validation methodology enables rigorous testing of these hypotheses and provides confidence bounds for the practical utility of the engineered features in ESP monitoring applications. Given the dataset limitations, particularly the scarcity of complete pump lifecycles, the approach focuses on identifying degradation indicators that can function effectively with partial lifecycle observations.

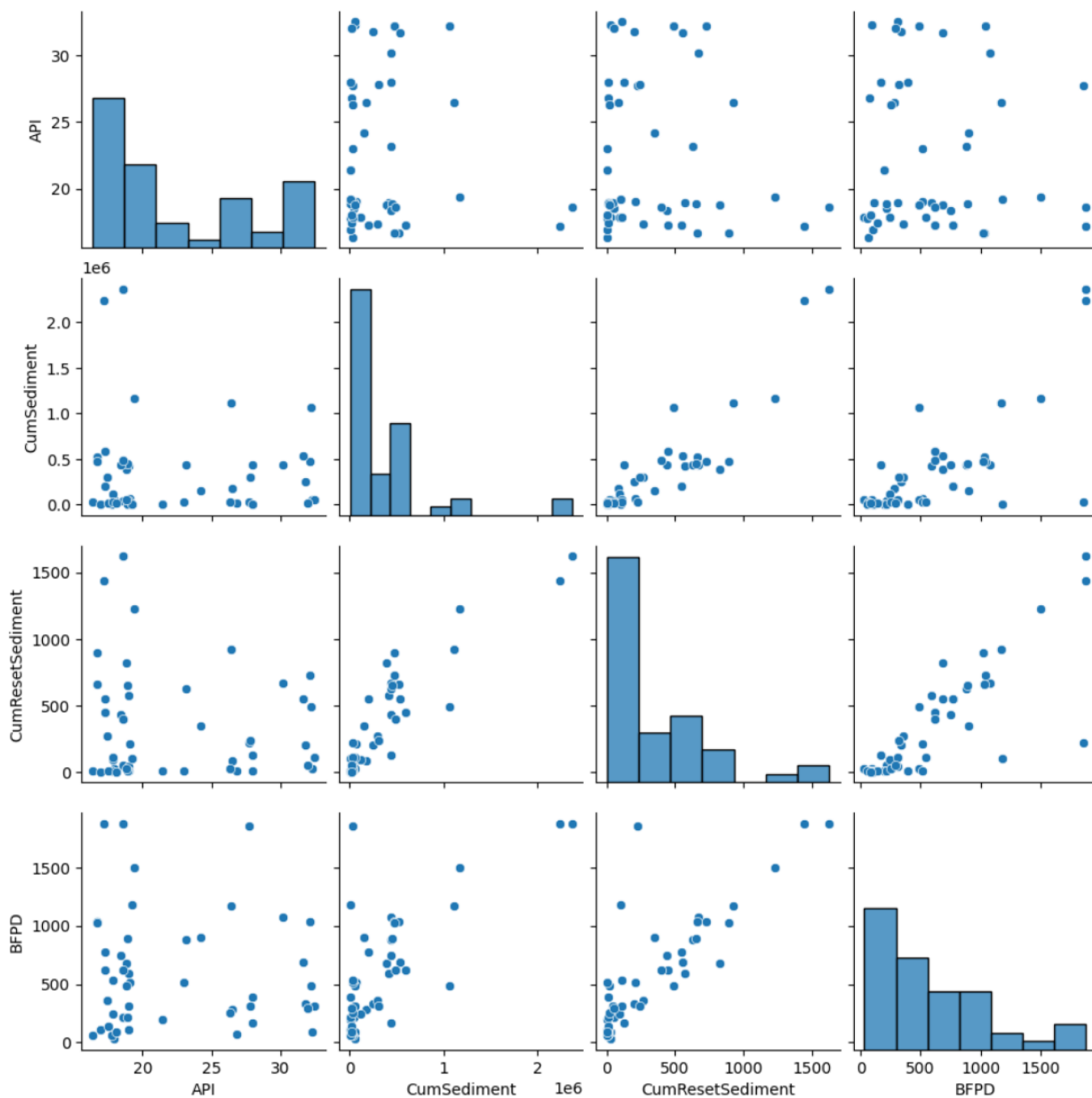


Figure 4—The figure shows a scatterplot for four chosen variables within the dataset at the time of failure. These four variables are among several assessed for gauging their significance in governing the survivability of the ESPs. CumSediment is the cumulative sediment at the time of failure using Equation 8. CumResetSediment is the cumulative sediment that is reset to zero every time a pump undergoes maintenance as specified in the original dataset. BFPD is the barrels of fluid per day normalized to 60Hz and API is the oil gravity.

Results and Analysis

We first generate the survival curve for the pumps in the entire field using Equation 3. Figure 5 shows the survival curve for the entire field. The median survival time is 232 days for the pumps in the field.

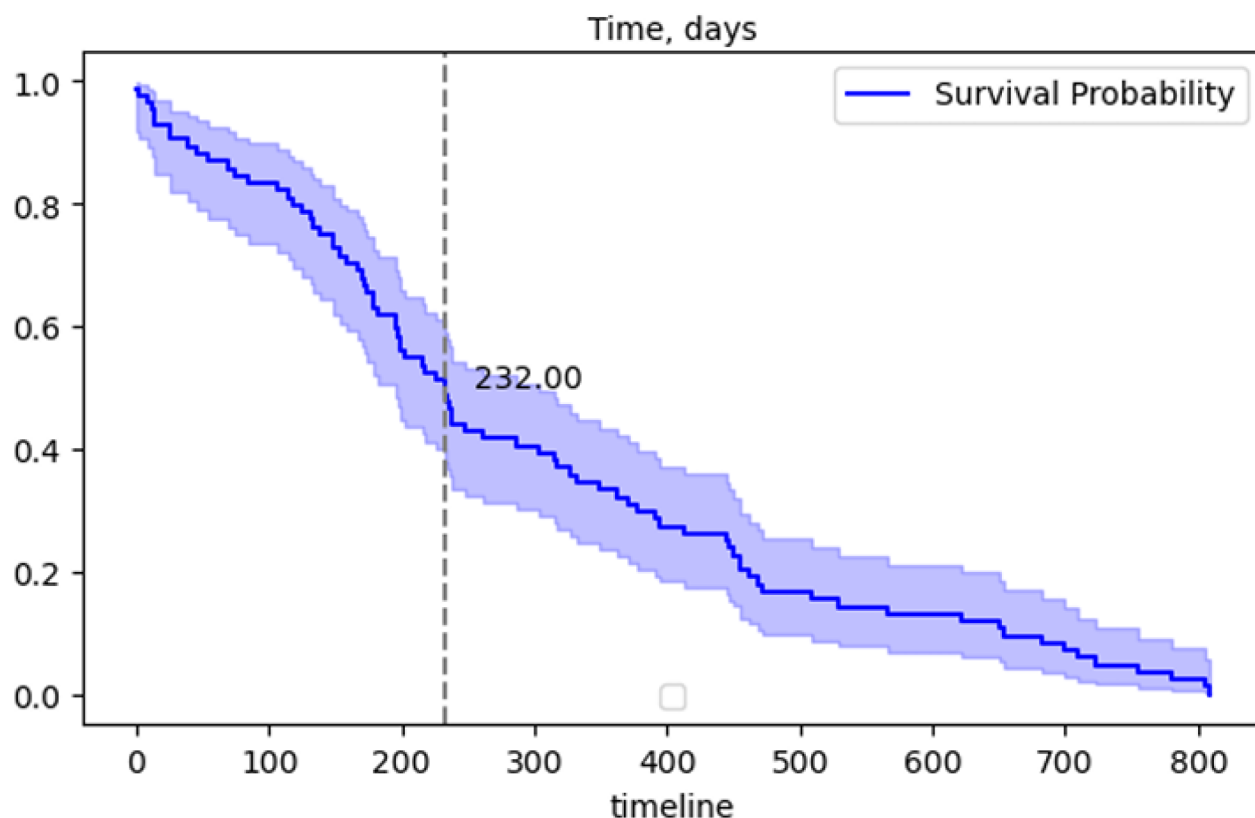


Figure 5—Survival probability curves for the pumps within the dataset.

Finally, given all the engineered features, we conducted a Log-Rank test to assess the impact of various variables in the dataset on the failure rates or survivability of the ESP pumps in the field. Before we perform the analyses, we first convert the continuous variables in the dataset to categorical variables with three levels (LOW, MEDIUM and HIGH). This is done by considering the lowest 25th percentile of each variable as ‘LOW’, the middle 50th percentile as ‘MEDIUM’ and the top 25th percentile as ‘HIGH’.

While this approach is not necessary, it gives us a better picture of assessing whether failure rates or survivability are different for different ranges of each variable in the dataset. For instance, we may be interested in knowing if the survivability of the pumps are different when the BFPD (barrels of fluid per day) is LOW versus HIGH or MEDIUM. Likewise, we may be interested in knowing if the oil API gravity controls pump survivability with any significance.

Figure 6 using the Log-Rank test shows that gas flows and the frequency-normalized flow rate, BFPD are significant variables that govern pump survivability. This observation corresponds to what is known about pump performance when gas production is substantial. A lesser-known effect is the effect of the frequency-normalized flowrate. This variable is a key indicator variable for ESP health because higher frequencies are associated with pumps struggling to deliver the specified flow rates for a variety of reasons, including a high gas fraction, increased sediment and others.

When applying the Cox Proportional Hazards model, we obtain similar results as shown in Figure 7. However, there are a few differences. The significant variables are BFPD (barrels of fluid per day) and LOW values of the gas flow rate (MSCF) and the variable Cumulative Reset Sediment (sediment values that are reset to zero when pump undergoes maintenance). This is also the rationale for creating categorical variables from the original continuous variables and dividing the range of each variable into LOW, MEDIUM and HIGH values.

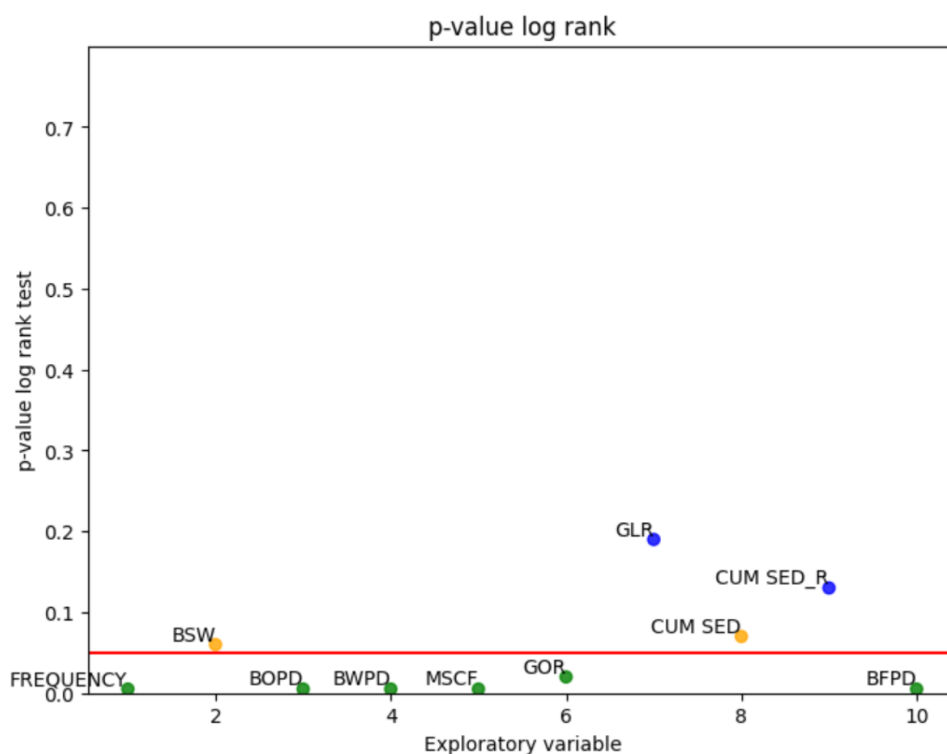


Figure 6—The results of the Log-Rank Test applied to the Ecuador dataset. Variables with a p-value of less than 0.05 are considered to be significant and explain pump survival. Gas production clearly influences pump survivability through the variables of GOR and MSCF. The frequency-normalized flow rate, BWFD is also a strong indicator of pump survivability.

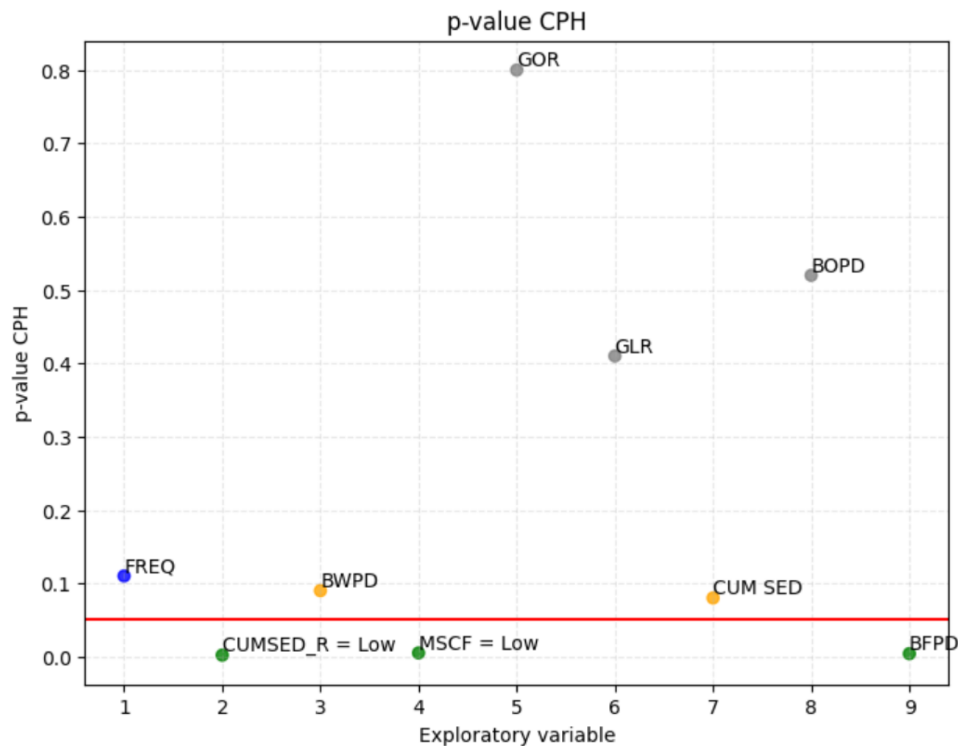


Figure 7—The results of the Cox Proportional Hazards Model applied to the Ecuador dataset. Variables with a p-value of less than 0.05 are considered to be significant and explain pump survival. Gas production clearly influences pump survivability through LOW values of MSCF. We observe the same with the Cumulative Sediment Reset value of LOW. The frequency-normalized flow rate, BFPD is also a strong indicator of pump survivability.

In spite of the differences between the outcomes of the log-rank test and the Cox Proportional Hazards model, it is important to note that both methods recognize the engineered features as significant. This likely would not have been possible had we utilized our original variables for analysis. This underscores the relevance of subject matter expertise creation and testing of various engineered features for gauging equipment survivability.

Conclusions

This paper describes and documents the use of remaining-useful-life prediction methods on a field case study. The following conclusions can be drawn from this study:

1. This study assesses the use of non-parametric and semi-parametric survival analysis methods regarding the failure of ESPs using field data
2. Survival analyses allow for meaningful interpretation of the median life of equipment and to identify the root causes of failure.
3. In spite of the limited information available, we were able to determine that gas flow rate and the frequency-normalized flow rate are critical parameters dictating pump survivability.
4. The use of appropriate feature engineering is critical. In our study, the fluid flowrate denoted by BFPD was not seen to be a critical feature but when normalized to a frequency of 60 Hz, we see a huge dependence of pump survivability on fluid flow rate. This is an insight that would not have been possible without some subject matter expertise.

References

- Aalen, O. O., Borgan, Ø., & Gjessing, H. (2008). *Survival and Event History Analysis: A Process Point of View* (First, Vol. 1). Springer New York. <https://doi.org/10.1007/978-0-387-68560-1>
- Akaike, H. (1973). Information Theory and an Extension of the Maximum Likelihood Principle. *International Symposium on Information Theory*, 267–281. https://doi.org/10.1007/978-1-4612-0919-5_38
- Al-Jazzaf, A. M., Pandit, A., & Al-Maqseed, N. (2019). Interpretation of ESP Reliability using Weibull Analysis and Dynamic Run Life Analysis. *SPE Gulf Coast Section Electric Submersible Pumps Symposium*, 2. <https://doi.org/10.2118/194393-MS>
- Bailey, W. J., Weir-Jones, I., Couet, B., & Hogan, J. R. (2006). Survival Analysis: The Statistically Rigorous Method for Analyzing Electrical Submersible Pump System Performance. *SPE Production & Operations*, 21(04), 492–504. <https://doi.org/10.2118/96722-PA>
- Brookmeyer, R., & Crowley, J. (1982). A Confidence Interval for the Median Survival Time. *Biometrics*, 38(1), 29. <https://doi.org/10.2307/2530286>
- Camacho, A., Pastre, L. F., Boyce, S., & Macdonald, J. (2016, November 30). High-Reliability ESP Application in the North Sea. *SPE Middle East Artificial Lift Conference and Exhibition*. <https://doi.org/10.2118/184171-MS>
- Camilleri, L., Hamza, L., Yucel, O., Rodet, V., & Breton, B. (2012, October 8). Lessons Learned from an ESP Development of a Naturally Fractured Carbonate Reservoir Offshore Gabon. *SPE Annual Technical Conference and Exhibition*. <https://doi.org/10.2118/159881-MS>
- Cantor, A. B. (2003). *SAS Survival Analysis Techniques for Medical Research* (Second, Vol. 2). SAS Institute Inc.
- Cirilo, R. (1998). *Air-Water Flow Through Electric Submersible Pumps* [M.S. Thesis]. The University of Tulsa.
- Cox, D. R. (1972). Regression Models and Life-Tables. *Journal of the Royal Statistical Society*, 34(2), 187–220.
- Divine, D. L., Lannom, R. W., & Johnson, R. A. (1993). Determining Pump Wear and Remaining Life From Electric Submersible Pump Test Curves. *SPE Production & Facilities*, 8(03), 217–221. <https://doi.org/10.2118/22399-PA>
- Fakher, S., Khlaifat, A., Hossain, M. E., & Nameer, H. (2021). Rigorous review of electrical submersible pump failure mechanisms and their mitigation measures. *Journal of Petroleum Exploration and Production Technology*, 11(10), 3799–3814. <https://doi.org/10.1007/s13202-021-01271-6>
- Greenwood, M. (1926). A Report on the Natural Duration of Cancer. *Reports on Public Health and Medical Subjects*, 33, 1–26.
- Hosmer, D. W., Lemeshow, Stanley., & May, Susanne. (2008). *Applied survival analysis : regression modeling of time-to-event data* (Second, Vol. 2). Wiley-Interscience.
- Hougaard, P. (2000). *Analysis of Multivariate Survival Data* (First, Vol. 1). Springer New York.

- Kaplan, E. L., & Meier, P. (1958). Nonparametric Estimation from Incomplete Observations. *Journal of the American Statistical Association*, **53**(282), 457–481. <https://doi.org/10.2307/2281868>
- Kartsonaki, C. (2016). Survival analysis. *Diagnostic Histopathology*, **22**(7), 263–270. <https://doi.org/10.1016/j.mpdhp.2016.06.005>
- Khabibullin, R. A., Shabonas, A. R., Gurbatov, N. S., & Timonov, A. V. (2020, October 26). Prediction of ESPs Failure Using ML at Western Siberia Oilfields with Large Number of Wells. SPE Russian Petroleum Technology Conference. <https://doi.org/10.2118/201881-MS>
- Klein, J. P., & Moeschberger, M. L. (2003). *Survival Analysis: Techniques for Censored and Truncated Data* (2nd ed.). Springer New York. <https://doi.org/10.1007/b97377>
- Kleinbaum, D. G., & Klein, M. (2012). *Survival Analysis: A Self-Learning Text, Third Edition* (Third, Vol. 3). Springer New York.
- Lewis, A., Audrey, D., Mercer, R. Jayaram, V. (2025). Case Study: AI-Driven Optimization of SWD Pumping Efficiency Using KPI-Based Time Series Analytics. SPE JPT July 2025, <https://jpt.spe.org/case-study-ai-driven-optimization-of-swd-pumping-efficiency-using-kpi-based-time-series-analytics>.
- Marques de Sa, J. (2007). *Applied statistics using SPSS, STATISTICA, MATLAB and R: Vol. Second* (Second). Springer Berlin.
- Mogollon, M. ., Arguelles, A. ., Rodriguez, A. ., Anaya, O. ., Miranda, S. ., Velasquez, E. ., & Villalobos, J.. (2018, March 13). Lessons Learned from Extending Run Life for Hundreds of ESPs in a Heavy Oil Environment. SPE Canada Heavy Oil Technical Conference. <https://doi.org/10.2118/189736-MS>
- Nguyen, T. (2020). *Artificial Lift Methods*. Springer International Publishing. <https://doi.org/https://doi.org/10.1007/978-3-030-40720-9>
- Pastre, L. F., & Fastovets, A. (2017). The Evolution of ESP Technology in the North Sea: A Reliability Study Based on Historical Data and Survival Analysis. *SPE Russian Petroleum Technology Conference*, 16–18. <https://doi.org/10.2118/187735-MS>
- Pessoa, R., & Prado, M. (2001, September 30). Experimental Investigation of Two-Phase Flow Performance of Electrical Submersible Pump Stages. Annual Technical Conference and Exhibition. <https://doi.org/10.2118/71552-MS>
- Sestelo, M. (2017). A short course on Survival Analysis applied to the Financial Industry: Vol. 1.2. *Bookdown. A short course on Survival Analysis applied to the Financial Industry*
- Sutyryna, O., Ovchinnikov, S., Yuldasheva, A., Lyzhin, B., Sofronov, I., & Pico, Y. (2022, November 15). A Digital Analytics Solution for Analyzing a Population of Electric Submersible Pumps. SPE Annual Caspian Technical Conference. <https://doi.org/10.2118/212135-ms>
- SPE Ecuador, (2021). <https://gitlab.com/spe.ec/e-challenge-prediction-models-for-historical-data-of-esp-patterns>
- VanderPlas, J. (2017). *Python Data Science Handbook* (Vol. 1). O'Reilly Media.
- Villalobos Leon, J., Correal, A., Pancho, J., Reyes, C., Fun Sang, B., Enriquez, L., Gallegos, J., & Torres, R. (2018). Data Management for Appropriate ESP Selection, Procedures and Operation Results in a Significant Reduction of Early Failures. SPE Middle East Artificial Lift Conference and Exhibition, 2. <https://doi.org/10.2118/192507-MS>
- von Flatern, R. (2015). The Defining Series: Electrical Submersible Pumps. *In Oilfield Review Defining Series*. SLB.